

IS115: ALGORITHMS & PROGRAMMING

WEEK 1B: COUNTING (COMBINATORICS)

Road Map

Algorithm Design and Analysis

- ♦ Week 1: Introduction, Counting
- ♦ Week 2: Complexity
- ♦ Week 3: Iteration & Decomposition
- ♦ Week 4: Recursion

YOU ARE
HERE

Fundamental Data Structures

(Weeks 5 - 10)

Computational Intractability & Heuristic Reasoning

(Weeks 11 - 13)

Counting

Basic principles in combinatorics



1. Product Rule
2. Sum Rule
3. Permutation
4. Combination

References

1. Rosen Chapter 6 (chapter from SMU library in eLearn)
 - 6.1: Basics of Counting
 - 6.3: Permutations & Combinations
2. Easy to read: <https://www.mathsisfun.com/combinatorics/combinations-permutations.html>

Scope:

Basic principles of counting:

1. Product Rule
2. Sum Rule
3. Inclusion-Exclusion Principle
4. Permutation
5. Combination

1. Product Rule

If there are m ways to do one thing,
and n ways to do another thing, then
there are $m \times n$ ways of doing both things.

E.g. 1 (product rule)

You need to do 3 projects in all: 1 from the industry, 1 from a prof, & the last project is to be self-proposed.

There are 5 projects offered by industry & 4 projects offered by profs. How many ways can you choose to do your 3 projects?

3 things to do:

- Pick an industry project - 5 ways AND
- Pick a professor's project - 4 ways AND
- Pick your own project - 1 way

Applying product rule, total = $5 \times 4 \times 1 = 20$ possible ways

E.g. 2 (product rule)

You need to meet your team for a project once every day on Mon, Tues and Wed (i.e. 3 meetings in total). On each day, there are 2 time slots that you can meet (1pm or 2pm).

These are the possible ways to arrange your meetings:

- Mon 1pm, Tue 1pm, Wed 1pm.
- Mon 1pm, Tue 1pm, Wed 2pm.
- Mon 1pm, Tue 2pm, Wed 1pm
- Mon 1pm, Tue 2pm, Wed 2pm
- ...

In how many ways can you arrange the meetings?

3 things to do:

- Pick a slot on Mon - 2 ways AND
- Pick a slot on Tue - 2 ways AND
- Pick a slot on Wed - 2 ways

Applying product rule, total = $2 \times 2 \times 2 = 2^3 = 8$ ways

E.g. 3 (product rule)

A binary string consists only of 0s and 1s.

How many possible strings of length n may exist?

- A binary string of length n has n binary characters (or bits).
- For illustration, let's assume $n = 3$. These are all the possible binary strings of length 3:
000, 001, 010, 011, 100, 101, 110, 111
- Each bit can be either 0 or 1, i.e., two possibilities.
- For 3 bits, we have 3 things to do together:
 - pick 1st bit: 2 ways
 - pick 2nd bit: 2 ways
 - pick 3rd bit: 2 ways
- Applying product rule, total = $2 \times 2 \times 2 = 2^3 = 8$
- Hence, for n bits, there are 2^n possibilities

2. Sum Rule

If there are m ways to do one thing,
and n ways to do another thing, and
both things cannot be done at the same time,
then there are $m + n$ ways of doing only one of the things.

E.g. 1 (sum rule)

You need to do 1 project only. You can choose a project from the industry, a prof's project or self-proposed one.

There are 5 projects offered by industry & 4 projects offered by profs. How many ways can you choose to do your project?

You can choose only one project.

- Pick an industry project - 5 ways OR
- Pick a professor's project - 4 ways OR
- Pick your own project - 1 way

Total = $5 + 4 + 1 = 10$ possible projects

E.g. 2 (sum rule)

You need to meet your team for a project one time only. You can choose either Mon, Tue or Wed. On each day, there are 2 time slots that you can meet (1pm or 2pm).

These are the possible ways to arrange your meetings:

- Mon 1pm,
- Tue 1pm,
- Wed 1pm.
- Mon 2pm,
- Tue 2pm....

In how many ways can you arrange the meetings?

You can choose only 1 slot:

- Pick a slot on Mon - 2 ways OR
- Pick a slot on Tue - 2 ways OR
- Pick a slot on Wed - 2 ways

Based on Sum Rule, total: $2 + 2 + 2 = 6$ ways

E.g. 3 (sum rule)

There are 2^3 (or 8) possible binary strings of length 3.

There are 2^4 (or 16) possible binary strings of length 4.

There are 2^5 (or 32) possible binary strings of length 5.

You have to choose only 1 binary string of length 3 or 4 or 5.

How many choices do you have?

You can choose only 1 binary string:

- Pick a 3-bit string - 8 ways OR
- Pick a 4-bit string - 16 ways OR
- Pick a 5-bit string - 32 ways

Based on Sum Rule, total: $8 + 16 + 32 = 56$ ways

Product Rule + Sum Rule

- More complex counting problems require both rules

E.g. 1 (sum + product rule)

You are required to complete a **two-phase** program:

- (a) Phase 1 (Research):** You must complete **1** project. You can choose either a project from a professor or a self-proposed project.
- (b) Phase 2 (Practical):** You must complete **1** industry project.

There are **5** industry projects, **4** professor projects, and **1** self-proposed option available. How many ways can you complete this program?

For Phase 1:

- no. of ways to choose a research project = $4 + 1 = 5$ (sum rule)

For Phase 2:

- no. of ways to choose a practical project = 5

Tot no. of ways to complete both phases = $5 \times 5 = 25$ (product rule)

E.g. 2 (sum + product rule)

A password must contain only capital letters (A-Z).
How many ways can you create a password of length 1-5
(inclusive)?

Applying the product rule:

- No. of ways to create a 1-letter password = 26^1
- No. of ways to create a 2-letter password = 26^2 26 $\times$ 26
- No. of ways to create a 3-letter password = 26^3
- No. of ways to create a 4-letter password = 26^4
- No. of ways to create a 5-letter password = 26^5

Applying the sum rule:

$$\begin{aligned} \text{Tot no. of ways to choose a password} &= 26^1 + 26^2 + 26^3 + 26^4 + 26^5 \\ &= 12,356,630 \end{aligned}$$

3. Inclusion-Exclusion Principle

If two things can be done at the same time, the sum rule will lead to double counting.

In this case, we add the number of ways to do each task, and then subtract the number of ways to do both tasks.

E.g. 1 (Inclusion-Exclusion Principle)

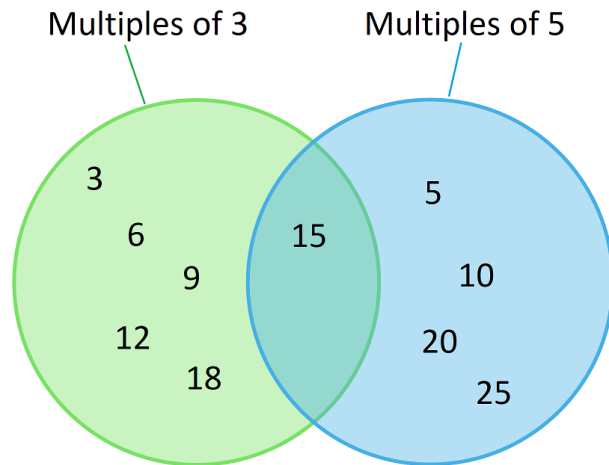
How many integers from 1 to 1000 are multiples of 3 or multiples of 5?

- Multiples of 3: {3, 6, 9, 12, 15, ... 27, 30, 33..., 999}
 - no of integers = $1000/3 = 333$
- Multiples of 5: {5, 10, 15, ...25, 30, 35..., 1000}
 - no of integers = $1000/5 = 200$
- $(333 + 200)$ is not the correct answer because multiples of both 3 and 5 (e.g., 15, 30) are counted twice. We must take the double-counting into consideration.
- Multiples of both 3 and 5: {15, 30, 45, ... }
 - no of integers = $1000/15 = 66$

66 numbers have been double-counted.
- Apply Inclusion-Exclusion Principle: total = $333 + 200 - 66 = 467$

E.g. 1 (Inclusion-Exclusion Principle)

How many integers from 1 to 1000 are multiples of 3 or multiples of 5?



- 333 integers in **green** circle (of which 66 are in overlapping area)
- 200 integers in **blue** circle (of which 66 are in overlapping area)
- 66 integers in overlapping area (double counted)
- Total number of integers in **green** and **blue** area

$$= 333 + 200 - 66$$

$$= 467$$

ICE for Sum/Product Rule + Inclusion Exclusion Principle

Q1.

Every car registered in Singapore has a license plate number, which begins with a letter 'S', followed by two other letters, 4 digits, and another letter (all upper case)

How many possible license plate numbers are there?

Assume that the last letter is not a checksum, & is independent from the other parts of the number. For example, there can be 2 cars having these plates: SAB1234X and SAB1234Y.

Q2.

It turns out that all plate numbers beginning with SH are for taxis only. How many possible license plate numbers are there for non-taxis?

Q3.

You are supposed to create a new password.

- Min 6 characters and max 8 characters.
- Alphanumeric characters (letters & numbers).
- Case insensitive (i.e. 'A' is the same as 'a')
- The password must contain at least 1 digit.

How many possible passwords are there?

Q4.

If passwords may only contain lower case letters and digits, how many 6-character passwords start with a lower case letter 'a' or ends with a lower case letter 'z'?

4. Permutation

A permutation of a set of objects is an ordered selection of these objects.

- Given a set $S = \{A, B, C\}$, its permutations are:
 - A, B, C
 - A, C, B
 - B, A, C
 - B, C, A
 - C, A, B
 - C, B, A

The number of permutations of a set of length n is $n!$

r-Permutation

An r-permutation of a set of n objects is an ordered selection of r of the n objects.

- Given a set $S = \{A, B, C\}$, its 2-permutations are:
 - A, B
 - A, C
 - B, A
 - B, C
 - C, A
 - C, B

The number of r-permutations of a set of length n is:

$${}_n P_r = \frac{n!}{(n - r)!}$$

E.g. 1 (Permutation)

150 countries compete in the Olympics.

How many ways are there to select 1st place, 2nd place, and 3rd place winners?

- **Solution 1:** Reason it out:
 - 1st place: 150 choices
 - 2nd place: 149 choices
(because 1st place cannot also be 2nd place)
 - 3rd place: 148 choices
(because 3rd place cannot also be in 1st & 2nd place)
- Total = $150 \times 149 \times 148$
= 3,307,800 ways

- **Solution 2:** Use the formula:
$${}^{150}P_3 = 150! / (150 - 3)!$$
$$= 150! / 147!$$
$$= 148 \times 149 \times 150$$
$$= 3,307,800 \text{ ways}$$

5. Combination

A combination of a set of objects is an unordered selection of these objects.

- Given a set $S = \{A, B, C\}$, its combination is:
 - $\{A, B, C\}$

The number of combinations of a set of length n is 1.

r-Combination

An r-combination of a set of n objects is an unordered selection of r objects out of the n objects.

- Given a set $S = \{A, B, C\}$, its 2-combinations are
 - A, B
 - A, C
 - B, C

The number of r-combinations of a set of length n is:

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

E.g. 1 (Combination)

150 countries compete in the Olympics. How many ways to select the top 3 winners if we do not care about their specific ranking?

- **Solution 1:**

- No of 3-permutations

$$= 150 \times 149 \times 148$$

$$= 3,307,800$$

- These 3-permutations contain all ways to order a selection of 3 countries. No. of ways to order 3 countries is $3! = 6$.
- No. of ways to select 3 countries =
 $3,307,800 / 6 = 551,300$.

- **Solution 2:** Use the formula:

$${}^{150}C_3 = 150! / 3! (150 - 3)!$$

$$= 150! / 3! 147!$$

$$= 148 \times 149 \times 150 / 6$$

$$= 551,300 \text{ ways}$$

ICE for Permutations & Combinations

Q5.

- i. How many permutations of the letters A to H if the 3 letters "ABC" have to occur consecutively?
- ii. What if they must appear together, but not necessarily consecutively?

ICE for Permutations & Combinations

Q6.

You are inviting 3 other guests (so there are 4 diners, including yourself) to have dinner at home.

Your rectangular dining table can seat 6.

How many different seating arrangements can you have?

Q7.

We would like to form a 7-member committee, consisting of 3 faculty members and 4 students. There are 40 faculty members and 300 students.

How many different ways can we form the committee?

Q8.

How many ways are there for 3 men and 3 women to stand in a line so that no two women stand next to each other?

Summary

Basic principles of counting:

- **Product Rule**: m ways to do one thing; n ways to do another thing; $m \times n$ ways of doing both things.
- **Sum Rule**: if only one thing can be done, $m + n$ ways
 - **Inclusion-Exclusion Principle**: if there are cases when both things are done together, we need to subtract the number of ways to do both from the sum.
- **Permutation**: number of ways to get an ordered selection from a set
- **Combination**: number of ways to get an unordered selection from a set

Press down 4
numbers to unlock



Each rotating disc has 10 digits:
0, 1, 2... 9



If the passcode is only 4 digits,
in the worst case, how many ways do you have to try?

